Wiring Diagram v002 (WIP)

Dustie Mk2

aka 'The VHS Wood CNC'

NEMA 6-15
240VAC
240v R, S, PE
Strain Relief Grommet
DIN Rail
Terminal Block
R
S
PE
MCB
2P 15A
TYPE C
AC EMI
Filter
L
N
GND
A1
1/L1
3/L2
5/L3
13 NO
A2
AC
Contactor
2/T1
4/T2
6/T3
14 NO
Ground Bus Bar
PE
PSU · 120 W · 24 V
Mesa 7i76EU
PSU · 350 W · 48 V
Y1 Stepper, Y2 Stepper
PSU · 350 W · 48 V
X Stepper, Z Stepper
PSU · 350 W · 24 V
Mesa 7i76EU, Rotary Axis A, Field Power
V+
V-
PE
N
L
Lenovo PC · LinuxCNC
← DVI to HDMI →
USB Extension
Ethernet
HY-DIV268N-5A
RJ45
ETH
Mesa
7i76EU
FRAME
GROUND
Inside Cabinet
Outside Cabinet
Umbilical
U
V
W
GND
Wall Mounted
Variable Frequency Drive
R
S
T
P+
PR
U
V
W
GND
Breakout Box 1
Static Platform
aka 'The Table'
Breakout Box 2
Y Platform
aka 'The Gantry'
Breakout Box 3
X Platform
aka 'The Toolhead'
Through-wire (4,4)
Multiple Connectors (4,2,2,2)
Connectors (2,2)
Connector (2)
Multiple Connectors (4,2,2)
Y1 Drag Chain
X Drag Chain
Power and Signal bundles separated in drag chain
DB25 (23 signals)
DB25 (18 signals)
DB9 (7 signals)
Stepper Y1
Stepper Y2
Stepper X
Stepper Z
PNP
NO & NC
SHARK Touch Plate
Static Platform Sensors
Endstops: Y+, Y-
Home Sensors: Y1, Y2
Touch Plate
Y Platform Sensors
Endstops: X+, X-
Home Sensors: X
Z Platform Sensors
Endstops: Z+, Z-
Home Sensors: Z